

Designing for Rest: Rethinking Access for / from Chronic Illness

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Abstract

In this paper, we share a case study of using critical disability studies to design for chronic illness. Specifically, we draw from the Political/Relational model of disability to explore designs for rest. Through a 6-week design collaboration, we brought together different design perspectives and worked through tensions between utility-oriented product design approaches and critical disability approaches for access. Our process yielded three design strategies: 1) moving from designing a product to designing a provocation; 2) using mapping as a process of building collective understanding of the built and social environment; and 3) re-imagining institutional practices around access as a starting point for design. We end by unpacking tensions in our design process and sharing some reflections on how to critically design for access in HCI.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

disability, chronic illness, rest, access, co-design, crip theory, critical disability studies

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1 Introduction

Lived experiences of certain energy-limiting chronic illnesses, such as long Covid, chronic fatigue syndrome, fibromyalgia, among many others, are marked by fatigue, post-exertion malaise, and a frequent need for rest [21]. This stands in stark contrast to the pervasive ideals of productivity and efficiency embedded in our social practices and built environments. Many digital technologies, from self-tracking apps to smart city networks, reinforce these ideals, encouraging us to become “fitter, better, and more productive” [63]. However, in doing so, they are “actively excluding” people with disabilities and chronic illnesses, who move at different paces [19]. Despite this tension, rest has rarely been considered as a form of access, and discussions on how to design for rest remain largely absent.

In HCI, designs that center disability experiences have often taken the form of assistive technologies, or devices that support people with disabilities in performing tasks, improving independence, and enhancing their quality of life (e.g., [15]). However, critical disability scholars have cautioned that designing assistive technologies without addressing the socio-cultural contexts in which they operate can create “techno-fixes” that further enable the design of inaccessible spaces or systems [60]. Further, some have suggested that the very term “assistive technology” is embedded in ableist rhetoric that implies a distinct category of tools designed exclusively for people with “impairments” to better function in “normal” society [25]. Instead, critical disability scholars highlight the importance of considering the political, relational, and intersectional aspects of disability when considering access [22, 37]. These ideas have been applied to the design of digital technologies [7, 8, 31, 42, 62].

For people living with an energy-limiting chronic illness, resting is critical for managing health, thus deeply entangled with questions

of access. However, rest is both political and relational, entwined with intersectional experiences and axes of oppression, and differently afforded to different bodies [27, 53]. Thinking through rest thus invites an examination of what it means to design for access, and who deserves access.

In this paper, we share a case study on designing for rest, guided by the question: *How can we design to support embodied experiences of illness, such as fatigue and post-exertion malaise, while advocating for broader cultural changes to resist ableist practices that work against chronic illness experiences?* In our case study, A1, a design and HCI researcher who lives with a chronic illness, worked with a group of industrial design master's students (A2-A5) over a 6-week period to explore designs for chronic illness, with a focus on illness-related fatigue and energy-limiting experiences. We drew from critical disability literature, specifically, the Political/Relational Model of disability [37]. Our collaborative exploration yielded three sets of design activities that bridged utility-centered product design approaches with an examination of the politics of rest. In our discussion, we share some generative tensions that arose through our design process and unpack how HCI scholars and designers might engage with technologies when doing the important work of building access.

2 Related Work

This section provides an overview of HCI design projects that center chronic illness experiences, introduces important critical disability literature that set the groundwork for the case study, and highlights scholarship on rest that informed the project activities.

2.1 HCI and Chronic Illness

Design work that frames chronic illness through the lens of access is under-addressed in HCI research [40, 41]. While chronic illness experiences are diverse and wide-ranging, generally speaking, chronic illnesses have mostly been approached through the individualistic model of disability, which views disability and illness as a personal problem afflicting individual people, and the closely related medical model of disability, which frames disability and illness as problems in the body to be "fixed" or cured. Design interventions thus tend to focus on controlling the body through technologies that inform individual behavior, primarily in the realm of illness management [41]. This is seen in designs of self-tracking technologies [2, 45], such as wearable devices for pacing [30] or applications for symptom management [11, 56], educational tools for health literacy [5], platforms for communication or sharing information with medical providers [10, 49], and platforms for finding social support [20, 33].

While HCI work that approaches chronic illness through the lens of accessibility is less common, some projects explore designs for chronic illness beyond the individual and medical model. For example, Mack et al. share a framework for designing of digital technologies that center the chronically ill experience [41]; Janicki et al. present a navigational platform for people with chronic illness to find restorative spaces in public [35]; Hendren details a case of a smart crosswalk that adjusts to the temporalities of different people, pacing the built environment instead of bodies [26]. In addition, Homewood shares a case of repurposing self-tracking technology to "do less", de-emphasizing fitness as a goal to show how technologies

can center rest as an outcome [29]. These projects offer examples of how technologies can be designed to improve accessibility for chronic illness beyond managing symptoms or controlling the body.

In addition, critical approaches for engaging with disability and access are also growing in HCI. Scholars increasingly draw from critical disability scholarship and crip theory [43] to design assistive technologies [7, 9, 42, 62] and engage with the politics of access through design. For example, this has been done in the context of critiquing normative health practices [50], rethinking digital access for data visualizations [31], and reimagining worlds from crip perspectives through speculative designs [1], among others. We bring these critical framings to rethinking designs for access, specifically for/from chronic illness.

2.2 The Political/Relational Model of Disability

In our project, we draw from critical disability studies, specifically the Political/Relational Model of disability, to explore designs for rest that center chronic illness experiences. In *Feminist, Queer, Crip*, disability scholar Alison Kafer argues that the individualistic and medical models of disability depoliticize disability and overlook the complexity and dynamic experiences of disability, reducing it to static categories removed from social and cultural environments [37]. Instead, Kafer proposes the Political/Relational Model of disability, which highlights disability as events, actions, and encounters produced through interactions. She suggests that "disability is experienced in and through relationships" [37, p8], shifts over time and by context, and is affected by one's culture and environment. Kafer calls for "a crippled politics of access" that rethinks systems and social practices while holding space for experiences of illness and a desire to be cured.

Hamraie offers an example for how digital technologies might support access in political/relational ways through the case of digital access mapping [23]. Using an intersectional framework for spatial study, Hamraie argues that accessibility is not merely a matter of compliance with laws or building codes, and should not be reduced to physical modifications of space, but involves a deeper consideration of the lived experiences of disabled people within various environments, including their sociocultural and relational dynamics. In this framework, digital technologies are not only for utility, but a way to ask questions and challenging assumptions [23].

Other disability scholars have built on this model to draw out implications for technology and design practice. For example, in *Crip Technoscience Manifesto*, Hamraie and Fritsch emphasize disability as entangled with the "intersectional workings of power and privilege" [24]. They highlight technoscience as a means for political action, calling for resistance against compliance with normal embodiment and to "agitate against independence and productivity as requirements for existence" [24, p2]. We draw from these calls in our design explorations for rest to highlight the dynamic and intersectional experiences of disability and technologies beyond utility.

2.3 Rest

Rest is a shared human right and necessity for survival, transcending its medical significance to encompass broader social, cultural,

and ethical dimensions of care and well-being. Rest is particularly critical in managing many chronic illnesses, especially ones that have an energy-limiting component, such as the more commonly known chronic fatigue syndrome (ME/CFS) and Long COVID [21]. These conditions often require a delicate balance between physical activity and rest, where exceeding one's energy capacity can result in significant setbacks.

In *Rest is Resistance*, Tricia Hersey offers a powerful framework for rethinking rest as a radical act of care and liberation [27]. Hersey critiques the relentless productivity and hustle culture embedded in extractivist capitalist systems and calls for rest as a form of resistance, particularly for Black and other historically marginalized communities exploited for their labor. Rest is a way to reclaim one's humanity in a society structured by capitalism, white supremacy, and other oppressive systems [27].

This view of rest aligns with the Political/Relational Model of Disability and opens up discussions on the unequal access to rest across built, social, and cultural spaces. For example, in his work on anti-homeless design, Rosenberger shows various examples of hostile architecture explicitly designed to prevent people from lying down, with the goal of deterring unhoused populations [53], many of whom live with disabilities themselves [32]. Digital technologies can reinforce this unequal access to rest and resting infrastructure, as seen in the example of the “homeless encampment data dashboard” released by the City of Portland, which documents locations of “homeless encampments,” where unhoused populations sleep, to increase the ease of policing [48]. This underscores the critical role of HCI design in shaping who has access to rest and who is systematically denied it.

Finally, in HCI, efforts to push against the culture of productivity are seen in the growing work on slow technologies. These include design projects that repurpose productivity tools for alternative ends [29, 34, 38], designs for slowing down [18, 46, 47], projects that engage with different lived experiences of time [16, 52], and work that confronts ideals of productivity and usefulness in HCI [39]. We align our work with these projects in engaging with the topic of rest.

3 Design Process

In fall of 2023, A7 invited A1 to serve as a co-designer in her class titled “Design with Access in Mind.” Co-design is a participatory design approach that actively involves stakeholders in shaping the design process and outcomes, emphasizing shared decision-making and mutual goal-setting [55], often used in the context of designing accessible technologies [62, 66]. In the class, A1 collaborated with A2-A5, a team of Industrial Design master's students, to explore accessibility through the lens of chronic illness. Together, the team worked to establish mutual goals and theoretical framing, integrating A1's lived experience of illness. A1 contributed critical perspectives on access and technology, while A6 and A7, both experienced in design research and HCI, supported the project's development as academic mentors.

The project aimed to explore a critical disability design approach for chronic illness, drawing specifically from the Political/Relational Model of disability. As such, while we started under a co-design framework, the project quickly moved away from utility-focused

product design approaches to exploring design as a way of asking questions and challenging assumptions about access in the context of chronic illness [22, 37].

The co-design process took place over a six-week period and with IRB approval. The process was iterative and exploratory, evolving organically without pre-determined activities or artifacts. Through initial discussions, A1 identified rest as a possible focus due to her experiences of fatigue and post-exertion malaise, in which a range of symptoms, such as inflammation, cognitive dysfunction, and pain, become exacerbated following physical exertion. Managing an energy-limiting condition is complex, but, as a starting point, living with this condition means that A1 often needs to sit down, sometimes lie down, throughout the day, avoid rigorous exercise, which on bad days can include walking, and prioritize sleep. A2-A5 found rest to be an engaging and complex design challenge, offering a way to critically examine access through the lens of chronic illness.

We focused on our university campus as boundaries of exploration, examining both campus culture and campus as a physical space. Weekly meetings formed the structure of our collaboration, serving as a space for collective brainstorming and mutual reflection, during which A1 would share her experience of chronic illness while bringing in a critical perspective on access from disability scholarship. Following each meeting, A2-A5 would develop and sketch ideas based on these discussions, returning with iterations for the next meeting. Occasionally, A1 would also sketch out ideas alongside the student team. This dialogical and iterative process allowed the group to collaboratively refine design directions, tasks, and goals over time. Throughout, we documented our individual and collective reflections, conversations, and design iterations in a shared Figjam board (See figure 1 as an example). During the design process, we encountered many tensions and negotiations stemming from different design approaches, namely, a product-oriented and utility-centered approach common in industrial design and the political/relational approach from critical disability spaces, which shaped our design outcomes, detailed in section 4.

Importantly, we acknowledge that chronic illness experiences are diverse and wide-ranging, and our project draws from only one person's experience of illness. We also recognize our privilege as academics who have basic access to healthcare, campus amenities and resources, and the time and space to engage in this research. Nevertheless, we present our case study as a way to demonstrate an approach of using critical disability studies in design and design education to draw out lessons, strategies, and provocations that can be further expanded upon.

4 From Co-Design to Designing for Resistance

The process of our design collaboration was messy, non-linear, and exploratory, and did not yield a single “polished” outcome. We conducted several design activities, experimented with different ideas, and switched directions multiple times. Throughout, we negotiated tensions of different design goals and priorities. In three themes, we share our process of 1) moving from designing products for resting to using these artifacts as provocation to make visible the politics of rest; 2) mapping restful spaces on campus to develop a functional navigational tool while building collective resistance; 3) integrating

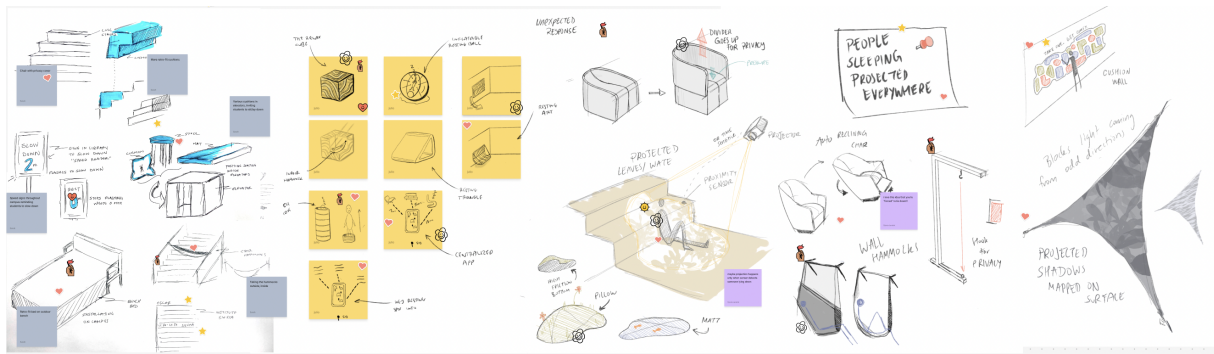


Figure 1: A screenshot of part of our collaborative Figma board shows preliminary sketches of ideas for designing for rest and comments and reactions from team members.

rest into institutional practices as an accessibility measure while observing tensions in this compliance-based approach. Under each theme, we discuss how the co-design team negotiated differences between designing for utility and engaging with the politics of access, and how the group made design decisions.

4.1 Moving beyond Restful Products

In this first set of outcomes, we share how we moved from designing restful products to positioning restful products as a form of provocation, and how this shift in design orientation was achieved.

The team's initial exploration of rest began with a focus on product design. In the beginning, A2-A5 focused on designing resting tools and amenities to support comfort, such as ergonomic cushions or napping pods. They gathered design inspiration images of different portable products for sitting down, lying down, and sleeping, and sketched out artifacts with various functions that supported different resting positions (see figure 1). Noting that these initial ideas primarily focused on utility and took the form of stand-alone objects, A1 encouraged A2-A5 to go beyond the objects themselves and explore the broader social and cultural implications of these designed things, prompting them with questions (e.g., where might these objects be used and under what situations? How would someone feel when using these tools in public spaces? Could these resting tools actually be working against the idea of rest that we want to convey? How might the use of napping pods in workplaces be a tool to fuel productivity?).

For the student team, the shift from designing products to designing “politically” or “relationally” was abstract at first, and they often came back to sketching specific objects. To catalyze a shift in our design direction, we stepped back from the drawing board and conducted fieldwork to explore current resting infrastructure and practices on campus, as outlined in the next section (4.2). We also gathered a different set of “inspiration” projects that were less product-oriented, and more provocative. Specifically, we looked toward speculative and artistic projects that have used resting artifacts to foreground the politics of and unequal access to rest. For example disabled artist Finnigan Shannon designed “Do You Want Us Here or Not,” an art installation featuring painted benches in galleries with provocative writing that critiques the inaccessibility of exhibition spaces [58]. They also designed “The Only Thing I

Love about Stairs,” in which she placed custom cushions over steps, “making them less functional as stairs and more functional as a place to sit” [59]. In addition, architect and disability activist Helen Stratford’s “Seats at the Table” consists of pop-up installations of public resting spaces that invite people to pause, rest, and lie down, featuring quotes from people living with invisible disabilities and chronic illnesses and gentle instructions for rest [64]. Finally, we drew inspiration from Sara Ross’ project “Archisuits”, in which she built custom cushions into a jogger suit that allows one to sit down or lie down on hostile architecture intended to deter resting: “The suits [...] allow a wearer to fit into, or onto, structures designed to deny them” [54]. These examples helped the co-design team concretely envision how design can move beyond the focus on the “object” to take on a performative role that can foreground intangible social and political issues.

Drawing on these examples, instead of designing tools solely to support comfort, we shifted our design direction to asking: *How can resting amenities, such as comfortable cushions, also be used as a form of resistance against ableist expectations? How can making something comfortable also make visible the intersectional and unequal access to rest?* Moving forward, A2-A5 brainstormed different ways that “comfortable objects” could also foreground the relational, political and intersectional aspects of rest.

One strategy we arrived at was to curate these objects in specific ways and contextualizing them in specific places. For example, instead of designing ergonomic cushions as stand-alone objects, one design explored the development of a participatory art installation in the form of a cushion sculpture that could be set-up in the library (figure 2), which has primarily hard surfaces for seating. The installation would encourage students in the library to grab cushions and make their spaces comfortable, while also inviting them to shape the collaborative cushion display. The co-designers also discussed placing the cushion installation in a library breakout room – quiet, private rooms intended for individual or group work, where “non-research activities” are explicitly prohibited. This placement would symbolically challenge conventional expectations of library spaces, redefining them as sites that not only facilitate productivity but also support care, rest, and alternative ways of being.

In a related exploration, we imagined an intervention for the central staircase of the design building on campus, where the student

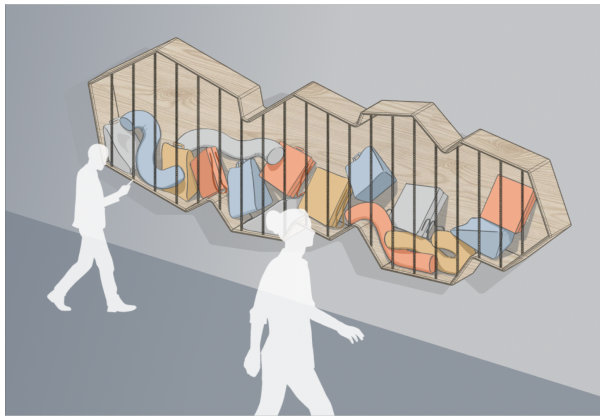


Figure 2: A cushion sculpture in the library would provide comfort while probing reflections on productivity culture.



Figure 3: Hammocks lining the main staircase of the Design Building foreground the implicit ableism in its dominance.

team's design studio was located. The staircase, the most prominent feature upon entering the brutalist style building (figure 3), is centered as a dominant architectural feature and the primary form of access to different floors in the building while elevators are hidden and difficult to find. In response, the student team proposed transforming the steps into a space for rest by hanging hammocks across the staircase - making visible the implicit ableism in its dominance.

While speculative, these negotiations helped shift the initial focus on designing objects that support rest purely from an ergonomic perspective to seeing design as provocation that makes visible the politics of rest. This shift encouraged the student team to situate their design interventions in context and in relation to specific practices in place, which allowed us to notice and address the different ways ableism was present in our everyday built and social environments.

4.2 Mapping Rest as Collective Resistance

In this second set of outcomes, we share our process of using mapping to understand the relational aspects of resting. All the while, the tension of moving away from product-oriented and utility-centered design approaches to engaging with the politics of access persisted. Drawing from Aimi Hamraie's concept of Mapping Access [23], A2-A5 conducted field work to document resting amenities, spaces, and practices around an area of campus (see figure 4). A1 encouraged the student team to consider the following questions when "mapping rest": Where are resting spaces and amenities? What are the physical and relational conditions around them? How do people use these spaces? Who has access, at what times? How do you feel when resting in these spaces? These guiding questions encouraged reflections on the relational aspects of resting beyond the physical infrastructure.

Following the fieldwork, A2-A5 organized their field notes and photographs, summarizing their observations and reflections on our shared Figma board. We then discussed these observations in a joint meeting, in which A1 also shared her personal experiences of finding resting amenities and restful spaces on campus. Our fieldwork and discussions reveal both physical and social barriers to resting on campus, highlighting a mismatch between the availability

of resting amenities and their accessibility and comfort. While there are many comfortable resting amenities, such as cushioned seating, throughout campus, there is rarely seating with headrests, and almost no places intended for lying down (with the exception of some outdoor hammocks). The student team further noted that even on seating where it was possible to lie down, they would not feel comfortable doing so because these spaces were often crowded, at times noisy, and lying down felt against the social norm, even if not explicitly prohibited.

The field observations also show how access to quiet and comfortable resting spaces on campus is often limited by spatial distribution, availability, and institutional rules, reflecting implicit and explicit norms about who can rest and where. For example, labs with restricted access, faculty lounges not intended for student use, and rooms that have to be reserved often have more comfortable amenities and atmospheres. While there are also some spaces designated for resting specifically intended for student use, such as a private "wellness" room in the library that could be reserved and a "relaxation room" in the student center that is open during building hours, these types of spaces are limited in number and across space. Some spaces are "policed" to prevent access by non-students while some buildings on campus have written rules prohibiting "sleep activity," lying down, and the use of pillows, blankets, sleeping bags.

In discussing these findings, A1 brought in some reflections on how intersectional experiences might also impact access to rest, and how these rules are entangled with social practices and cultural contexts that may privilege the comfort and safety of some people over others, depending on their positionality. For example, we discussed a case on a different US campus where a white student called the police on a black student who was napping in a common area in a campus building [17]. These intersectional experiences and cultural and relational specificities are important to questions of access [24, 27, 37, 43]. Recognizing we only represent a subset of the student population and collective experiences resting on campus, we discussed ideas to capture the experiences of students from different backgrounds, leading to an informal pop-up survey that invited other students to contribute their resting practices on campus (see figure 5).



Figure 4: The student team explored restful spaces on campus and documented their experiences.

The fieldwork yielded a collection of resting experiences that we organized on a shared map, which included multi-modal information about place, such as written annotations, photos, and videos, for the documentation of socio-spatial practices, following Hamraie [23] (figure 6 left). This map brings together the many resting amenities and their contextual information.

In compiling this information, we were met with another moment of tension between product design and relational/political approaches. The student team was eager to build out the map in the form of an app with its own user interface and interaction flow. A1 initially pushed against this idea, feeling it might distract from the political tone of the collective map with its product aesthetic, and preferred to emphasize the process of mapping and the collection of relevant information. In our discussions, we negotiated the potential barriers to access with possible benefits of designing a new app. For example, developing a stand-alone app as a crowd-sourcing tool requires users to make and manage a new account, whereas using an existing mapping platform that most people were familiar with, like a shared map list on Google Maps, would allow anyone to access and contribute information on a browser without downloading a new app or making new accounts. We settled on designing a simple UI as a mockup that streamlined the process of accessibility information collection with an intentionally designed UI to emphasize accessibility (figure 6 right).

In sum, the collective mapping process resulted in greater understanding of the social and built environment while allowing the design team to think through the relational conditions of rest. The map reflects a process of sharing knowledge about restful spaces that can be both functional and a start for working toward collective care and resistance.

4.3 Accessibility Measures for Rest

The final set of outcomes emerged from discussions on how rest might be institutionally recognized as an access need. This process came about following the previous mapping exercise, where we explored a bottom-up approach for accessing rest. Respectively, we also discussed the importance of institutional support for access. Writings on infrastructural speculations [67] helped inform our brainstorming process on how resting amenities and practices

might be integrated into campus' accessibility standards and accommodation practices for students with chronic illness, and subsequently, how these practices might be supported through the design of specific tools or technologies. Infrastructural speculations denote "an orientation towards speculation that aims to interrogate and ask questions about the broader lifeworld within which speculative artifacts sit," rather than an individual artifact [67], which aligned with our political/relational exploration of access.

As part of this brainstorming exercise, the team identified a list of potential stakeholders on campus who may have an interest in supporting rest as an accessibility measure, including student groups, academic centers, and other institutionally recognized interest groups. We then brainstormed ideas for integrating rest into campus operations speculatively. This process brought forth additional tensions of how a compliance-based approach for accessibility might reinforce ableist norms, create boundaries of exclusion, and risk commodifying rest. We share some ideas and discussions below.

First, we imagined a library checkout system that would lend out resting tools or devices to support the learning experiences of students with chronic illnesses. The integration of a check out system that provides "resting artifacts" could offer recognition of chronic illness needs while providing greater comfort. In brainstorming specific resting artifacts, A1 suggested that she sometimes wish she had a desk that could be used while lying down on days she felt more fatigued; the team also considered a detachable headrest and custom cushions that could make campus furniture more restful and comfortable, supporting students who experience fatigue. At the same time, the team discussed how this system might reinforce some ableist norms or fuel the culture of productivity. For instance, would we need detachable headrests if more chairs were comfortable? Would a lie-down desk push more work to be done instead of actual breaks and rest?

Next, we imagined how each building on campus could have a quiet resting space available for students with disabilities and chronic illnesses as an accessibility requirement. There are many existing models of public restorative spaces that provide access for different populations, such as lactation rooms, prayer rooms, sensory rooms, or even single stall gender-inclusive restrooms. We discussed how a simple stall with basic amenities may bring substantial relief to students who need a private space to rest and

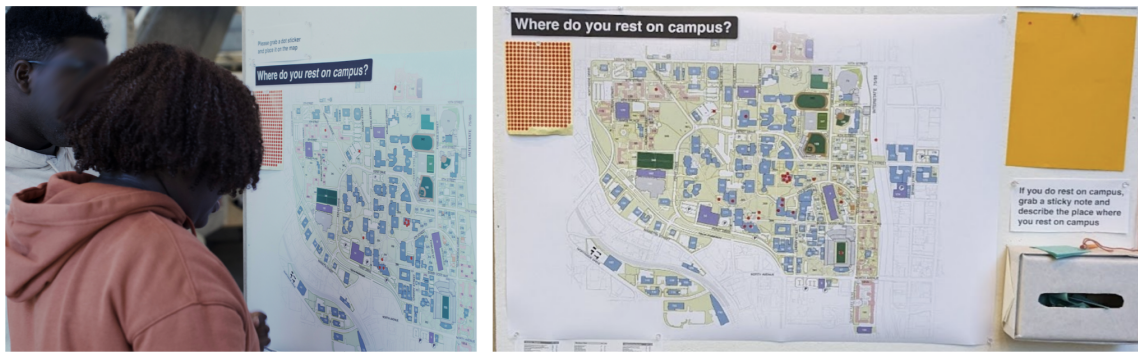


Figure 5: The student team set up an interactive pop-up survey to invite other students to share their experiences.

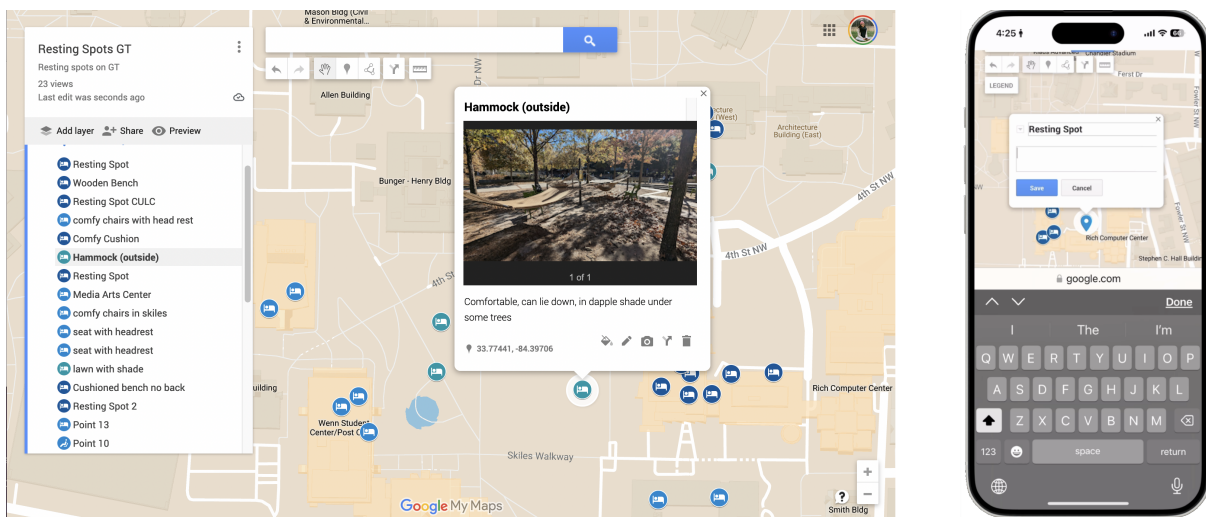


Figure 6: We organized findings from our fieldwork on resting spaces into a shared list in Google Maps (left); the student team then designed a mobile app prototype to streamline the process of information collection (right).

lie down during the day, and the importance of these spaces to be spatially distributed across campus to effectively support the access in the context of fatigue. We identified some underutilized spaces in our respective campus buildings and discussed some design ideas that could make the spaces comfortable and accessible. This led to some reflections on the tensions of inclusion and exclusion in these resting spaces: Who gets to have access? Whose bodies should be prioritized? Who decides? Who would fund, manage and care for the spaces? These considerations underscored the messiness accommodation practices that at times risk reinforcing fixed categories of disabilities and existing inequities.

Finally, we considered the idea of integrating rest into curricula, academic programs, or campus life. For example, could course credit be allocated for resting like they are for physical education classes? Should students be expected to have “offline” days? We unpacked the irony of a “resting class,” the problems of “policing” time off, and how these practices may contribute to ideals of productivity and control, which is particularly problematic in the history of how people with disabilities were treated by institutions [61]. Nonetheless, these reflections might be seen as small steps toward a cultural

shift that values rest equally with physical activity and productivity, which could support broader systemic changes over time.

In each of these scenarios, we began by speculating on small organizational changes, which sometimes invited the design of specific technologies. Our discussions underscore both possibilities and limitations of designing within institutional systems while resisting it — an approach that is not always radical, requires incremental changes, and can at times feel “unproductive”. Here, we also bring forth some tensions of using this speculative approach in the context of access work, especially under the Political/Relational Model of disability. Namely, as we “imagine” how rest may be integrated as a measure of accessibility in institutional spaces, it is important to recognize the critical labor involved in doing this political work on the ground. In practice, achieving institutional changes for access, even small changes, takes time, physical and emotional labor, and requires collective effort, usually with disabled people bearing the brunt of the labor [22]. In addition, as we “imagine” new systems of compliance for institutions, it is important to contextualize these in the history of crip resistance and *noncompliance* that lay the foundation for any legal and formal requirement for



Figure 7: The student team explored the design of different forms and shapes of cushions based on the types of seating available in the library, then modeled and rendered the designs in Fusion 360, Keyshot, and Photoshop.

accessibility. Our broader goal of this brainstorming process is to contextualize access within institutional systems and practices and foreground the inherently political work of access and design, as "design materializes politics" [22].

5 Discussion

In this paper, we shared a case study of using critical disability studies in the classroom setting to explore what designing for access might look like beyond the medical and individualist approaches of assistive technologies and the emphasis on utility. This builds on other HCI projects that have used crip theory in re-imagining access [1, 31, 50, 62] and responds to calls for using critical disability studies for teaching [42]. We explored how this critical design approach might apply to chronic illness through the topic of rest. In this section, we unpack three major tensions and related strategies that arose through the project, including: 1) negotiating product design and relational/political approaches for access, 2) centering disability experiences without reinforcing categories of disability, and 3) designing (with) technologies to shape social and political conditions for access.

5.1 Negotiating Product Design and Political/Relational Approaches for Access

Drawing from the Political/Relational Model of disability, our design process pushed against more utility-centered product design approaches to yield three strategies that supported a critical engagement with access: 1) moving from designing a product to designing a provocation, specifically, by positioning artifacts in relation to specific practices in place; 2) using mapping as a process of building collective understanding of the built and social environment instead of using maps only as functional tools; and 3) re-imagining and negotiating tensions of institutional practices around access as a starting point for design. These strategies helped us move away from dominant solutionist design approaches to engage with socio-cultural conditions in which designs operate, which is central to the Political/Relational Model of disability [37]. We share these as design directions that can be facilitated in classrooms to expand design possibilities for accessibility.

These strategies took form through many negotiations and discussions between members of the co-design team, who hold divergent design experiences and priorities. Throughout, we iteratively worked through the tensions of designing for utility, which is dominant in industrial design spaces that A2-A5 occupy, and a critical design approach that foregrounds the sociopolitical context of access. We brought together design examples from different perspectives, identified conflicting goals and values through conversation, and experimented with different design processes to share ideas and build collective understandings - from sketching individually on paper to collectively diagramming on Figma, from mapping the built environment to engaging with the broader campus community through pop-up surveys. We often reached design decisions by finding common ground, i.e. designing something tangible and functional while engaging with broader politics of access. For example, in 4.1, we both designed comfortable cushions and integrated them into site-specific installations as a form of provocation on the culture of productivity; in 4.2, we worked through how mapping can be both a functional tool and represent a form of collective resistance; in 4.3, we situated assistive resting tools as part of larger institutional changes for access. At times, we also negotiated whether to design tools that would support immediate physical needs but at the same time risk reinforcing productivity ideals or perpetuate techno-ableism [60], as shared in 4.3. These negotiations show how utility-oriented design approaches and critical design approaches can work together to address access needs. Bringing together interdisciplinary perspectives and working through differences brought these conversations to the fore, helping us to put designed objects in context, think through the messiness of access, and unpack the complex role of design in building access. This process also foregrounded "access as negotiation, rather than as a resolved, measurable end," as suggested by Hamraie [22].

5.2 Starting from, but Expanding beyond Chronic Illness

Next, we unpack the tensions of approaching access from specific experiences of disability while not reinforcing fixed categories of disability. Our design exploration began with A1's experience of chronic illness, which helped us identify rest as a form of access need. However, the design outcomes were expansive - extending beyond A1's chronic illness needs and inviting participation broadly.

We were mindful of designing for rest in ways that did not reinforce rigid categories of disability and illness, following crip framings of disability as fluid, dynamic, and intersectional [37]. For example, in 4.1, we imagined ways of integrating resting infrastructure in place that spoke to a broader culture of productivity and provided greater comfort for students; in 4.2, we invited students to contribute to mapping processes through pop-up surveys to expand beyond our own perspectives and positionalities; in 4.3, we imagined how incorporating rest as an accessibility measure for chronic illness might invite broader cultural shifts related to productivity ideals. In each case, we started by considering more specific or immediate accessibility needs in campus spaces for a student with chronic illness, i.e. physical amenities or accommodations, then expanded to questions of access that include broader social norms and political structures.

Disability scholar Sara Hendren suggests undoing the distinctions between “design for disability” and “design in general” [25]. She argues that “all technologies are assistive” because “we are all getting help from the things we make” [25, p141]. At the same time, Hamraie highlights the importance of centering disability experiences in designing for access, even as these efforts often lead to broader societal benefits. Through the case of curb cuts, Hamraie showed how universalist framings of access dilute the history and political significance of disability activism and frame accessibility as a byproduct of design rather than a hard-won right [22].

As we navigate this tension, we highlight the importance of access work to *start from* disability experiences, as reflected in our design exploration for rest, with recognition of disability identity, history, and community, and careful attention to hierarchies and power relations. In particular, we call for engaging with chronic illness experiences as a starting point for rethinking access in the built and social environment, responding to the under-addressed design space in HCI [35, 40, 41]. Chronic illness experiences, while diverse and wide-ranging, are deeply entangled with intersecting social and structural factors. Studies have shown that chronic illnesses unequally impact women [13, 65], racial minorities [51], unhoused populations [32], and low-income populations [36]. Thinking through access for chronic illness demands careful consideration of how disability is entangled with different axes of oppression, and invites an intersectional approach that not only addresses individual needs but also interrogates and transforms the broader social, economic, and structural conditions that perpetuate inequities.

5.3 The Role of Technology in Building Access

Finally, we invite HCI scholars to rethink the role of technologies in building access. We reiterate the importance of assistive technologies design and research in improving the lives of people with disabilities, especially within existing systems and structures. We also acknowledge that, as technology designers and researchers, HCI scholars are often tasked with building tangible products or technological solutions that are at times in conflict with broader critical goals. However, our work invites other ways of engaging with technologies to support access that can work alongside assistive and utility-centered approaches.

First, we highlight the potential of using technologies to build collective practices that can work toward cultural change. In our work, we explored digital mapping as an example of “organizing”

collective experiences that invite restful practices on campus. We point to other HCI projects that have demonstrated ways in which technology design and practices can be used to enact cultural or political change, such as Fox et al.’s work in *Hacking Culture, Not Devices* [14], and Bennett et al.’s work on interdependence as a frame for assistive technology research and design [7]. These works highlight the role of technological practices in building community and relations that create cultures of access.

Second, we share how technologies can be used to generate questions and debates around access, rather than a form of problem-solving [23, 25]. HCI projects have long engaged with design as a form of problem-setting and provocation, as seen in critical, reflective and speculative design traditions [3, 4, 12, 57, 67]. These speculative approaches can also be generative in the context of access. For example, we drew from infrastructural speculations to ask questions about what forms of institutional support for resting is possible on our campus. [67], building on speculative HCI projects that re-imagine access, such as Angelini et al.’s speculative stories that explore desirable worlds from a crip perspective [1] and Hsueh et al.’s speculative designs of crip-informed data visualizations [31]. These examples highlight the importance of imagining otherwise to move conversations about accessibility beyond the confines of existing systems.

Third, we advocate for a more humble view of digital technologies in the context of access, positioning them as supportive and supplementary tools that operate in tandem with built infrastructure and broader social, cultural, and political systems. This view challenges technosolutionist orientations by recognizing the limitations of digital technologies in certain contexts and acknowledging that they may not be necessary or may be harmful in others. In doing so, we emphasize the role of HCI scholars in negotiating and articulating these tensions in their work, and in making important decisions about when *not* to design [6, 28]. Through this, we join calls on decentering the innovation discourse in digital access design [31].

To conclude, we call attention to the need for a cultural shift toward valuing rest in HCI spaces. Recent scholarship has highlighted the culture of overwork in HCI [44] and a move toward designing to “slow down” [18, 46, 47]. As we share examples of designing access for chronic illness through the case of rest on our university campus, we also call for a re-examination of *how* we work by pointing to other HCI projects that have offered tactics of resistance against the norms of productivity [29, 34, 38, 39, 44] and invite reflections on how we might bring rest into our own academic practices as a start for building access in educational spaces.

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